

CAD/CAM Quadruped Robot – Assignment Problem Statement

Design a **simplified quadruped robot** and validate **basic leg gait motion** using CAD modeling, mathematical gait calculation, and simulation, focusing on correctness rather than complexity.

Questions to Be Solved

Question 1: Mechanical Structure Design

Design a **simplified quadruped robot** with:

- Rectangular body frame
- Four identical legs
- **2 DOF per leg** (hip and knee)

Task: Create a clear CAD model showing the body, legs, and joint locations.

Question 2: CAD Modeling and Assembly

Model all components using CAD software and assemble them into a complete quadruped.

Task: Generate a complete CAD assembly and demonstrate correct leg symmetry and joint alignment.

Question 3: Joint Definition and Motion Limits

Define joint rotation axes and reasonable motion limits for hip and knee joints.

Task: Specify joint limits and justify them based on mechanical feasibility.

Question 4: Motion Validation via Simulation

Validate leg motion using simulation tools.

Task: Simulate joint motion and verify smooth, collision-free leg movement.

Question 5: Gait Cycle Calculation

Compute gait parameters using basic kinematic relations.

Task: Calculate gait cycle time, phase offsets, and joint angle profiles for one leg.

Question 6: Stability Check

Perform a basic stability check.

Task: Demonstrate that the robot maintains static stability during walking using foot contact positions.

Question 7: Design Limitations

Analyze design constraints.

Task: Identify at least two mechanical or gait-related limitations of the current design.

Tools

Allowed

Preferred

- **SolidWorks** – CAD modeling and assembly
- **ANSYS (Motion / Structural)** – Simulation

Free Alternatives (Allowed)

- **Fusion 360 / FreeCAD** – CAD
- **GNU Octave / Python** – Gait calculation
- **SimScale** – Simulation

Submission Guidelines (Google Drive)

Upload all files to **Google Drive** and submit the shared folder link.

Folder Name

Quadruped_10Day_MiniProject_<StudentName>

Contents

- Project report (PDF)
- CAD files or STEP models
- Simulation screenshots or short video



INDUSTRIAL TRAINING PROGRAM ON ROBOTICS AND AI